Frontier LLMs Fall Outside Human Moral Space

A Replication and Extension of the Moral Machine Experiment Across Seven Frontier Systems

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The Moral Machine experiment collected over 40 million human judgments on autonomous vehicle dilemmas across 130 countries, establishing the largest empirical record of human moral preferences along nine dimensions. We replicate that paradigm on seven frontier large language models (LLMs) released in 2025, spanning six providers across the United States, Europe, and China, and comparing 27,300 model decisions against country-level human baselines using Average Marginal Causal Effects (AMCEs).

All seven models fall outside the convex hull of all 130 countries in the nine-dimensional AMCE space: no model's moral profile is matched by any human population, and 47 of 63 model-by-dimension comparisons differ significantly from the human average after FDR correction. Inter-model divergence substantially exceeds cross-country human variation; on the species dimension the range across models is roughly five times the entire cross-country human range. Two dimensions on which humans are consistent are near-universally suppressed: all seven models are effectively gender-neutral and status-neutral, against human preferences of $+0.12$ and $+0.35$, a pattern consistent with fairness interventions targeting stereotyped generation generalizing into moral reasoning that was never their target.

We further introduce a systematic prompt sensitivity analysis. Minor reformulations move no estimate significantly (0 of 63), while a single-sentence ethical override reverses the species preference from $+1.00$ to -1.00 in four of seven models, dividing systems into three response types: compliance, indifference, and resistance. These results locate the moral profile of a deployed LLM in provider training and alignment choices rather than in any human population, making provider selection a consequential and currently undisclosed moral decision, and they show that the resulting profile can be overridden by an adversary controlling one line of context.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Social and professional topics** → **Computing / technology policy**; • **Human-centered computing** → *Empirical studies in HCI*.

Additional Key Words and Phrases: large language models, moral preferences, Moral Machine experiment, value alignment, autonomous vehicles, prompt sensitivity, AI governance, model auditing

1 Introduction

As autonomous vehicles, surgical robots, and other safety-critical systems increasingly incorporate AI decision-making, the question of whether AI systems share human moral intuitions becomes practically consequential. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded, implicitly or explicitly, in its behaviour [7].

The Moral Machine experiment [2] provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle. The study identified nine binary moral dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate human preferences, quantified as Average Marginal Causal Effects (AMCEs), vary substantially across cultural clusters while sharing a common core. A controlled follow-up across 42 countries confirmed the robustness of these cross-cultural patterns [3]. The moral psychology of AI as an emerging field is surveyed in Bonnefon et al. [6].

The rapid deployment of LLMs in agentic and embodied settings [8, 9], including robotics platforms capable of physical action, makes the question of LLM moral alignment newly urgent. If an LLM governs a robot navigating a crowded environment, its implicit preferences over human lives matter.

The governance framing matters as much as the empirical one. When a moral preference ordering is encoded in a deployed system, three accountability questions follow: whose preferences are they, are they disclosed to the people subject to them, and can they be changed by someone other than the deployer. Existing transparency instruments are largely silent on all three. Model cards and system cards document capabilities, evaluations, and safety mitigations, but no major provider publishes the moral preference profile its post-training produces, and no procurement or certification regime requires it. A purchaser choosing between vendors for a safety-critical deployment currently has no disclosed basis on which to compare their moral behaviour, and no way to detect that the two vendors differ.

This paper supplies the missing measurement. It makes three contributions. First, it evaluates seven frontier models released in 2025, none previously studied in this context, selected to span the tier of publicly available frontier systems as of early 2026 across six providers and three geographies, including providers underrepresented in this literature: DeepSeek (China) and Mistral AI (Europe). Second, it introduces a systematic comparison of prompt variants, from minor reformulation to explicit ethical override, which separates the stability of a model's moral profile under incidental variation from its vulnerability to deliberate override. Third, it reads both results as an accountability problem: the divergence between models is large, invisible to the people affected by it, and reachable by an adversary with control of a single line of context.

2 Related Work

Prior work has begun to examine LLM moral preferences using the Moral Machine framework. Takemoto [18] first replicated the experiment on four models (GPT-3.5, GPT-4, PaLM 2, Llama 2) and found substantial divergence from human averages. Ahmad and Takemoto [1] scaled this to 52 models, finding that larger models tend to be closer to human preferences. Vida et al. [19] and Jin et al. [12] extended the paradigm to multilingual settings, finding intra-model variation across languages.

A parallel literature documents that model values are shaped by post-training rather than pretraining data alone. Santurkar et al. [17] show that fine-tuning choices, not corpus composition, drive whose opinions a model reflects, and Kirgis [13] find that variation across LLMs exceeds variation across human demographic groups. Work on alignment methods themselves — RLHF [16] and Constitutional AI [4] — describes the mechanisms by which those preferences are installed, while Gabriel [10] sets out the normative question of which preferences alignment ought to target. Studies of stereotype mitigation in generation [14, 20] document the interventions whose side effects we observe on the gender and social status dimensions.

Relative to this literature, our contribution is to pair a new generation of frontier models with an explicit manipulation condition, which prior Moral Machine replications do not include, and to report the resulting profiles as auditable, provider-attributable properties rather than as aggregate model behaviour.

3 Methods

3.1 Scenario Generation

This study replicated the Moral Machine scenario structure following Awad et al. [2]. Each session consists of 13 paired dilemmas presented in a fixed block. Across 100 sessions, 1,300 unique scenarios were generated, each represented as two rows encoding the following binary attributes: intervention type (stay/swerve), barrier presence, pedestrian composition (pedestrian-vs-pedestrian or pedestrian-vs-passenger), crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

3.2 Models

Seven frontier models were queried via the OpenRouter API, which provides a unified OpenAI-compatible interface (Table 1). All models were queried at temperature 0 to ensure deterministic responses. Each response was required to end with `STAY` or `SWERVE` on a final line; rows where no valid choice could be extracted were flagged as parse failures and retried. Gemini 3 Pro incurred 34 safety-filter parse failures across the three prompt variants (13 standard, 17 emphasis, 4 manipulation), all of which were excluded; these are distributed across scenario types without concentration on any single moral dimension. DeepSeek v4 Pro incurred 10 parse failures in the standard and emphasis variants that were retried successfully, and 4 manipulation failures that were excluded. In total, 27,300 API calls were issued. Full per-variant AMCE estimates, complete pairwise inter-model z-statistics, PCA eigenvector loadings, scenario counts, and the verbatim prompt templates are provided in the supplementary material.[1]

Table 1. Models evaluated in this study.

Model	Provider	OpenRouter ID
Claude Opus 4.7	Anthropic	`anthropic/claude-opus-4-7`
GPT-5.5	OpenAI	`openai/gpt-5.5`
Gemini 3 Pro	Google	`google/gemini-3.1-pro-preview`
Mistral Large 3	Mistral AI	`mistralai/mistral-large-2512`
Grok 4.1	xAI	`x-ai/grok-4.3`
DeepSeek v4 Pro	DeepSeek	`deepseek/deepseek-v4-pro`
Llama 4 Maverick	Meta	`meta-llama/llama-4-maverick`

3.3 Prompt Variants

Each model received two main prompt variants sharing the same system role (ethical advisor for autonomous vehicle policy) but differing in the closing instruction: **standard** ("Analyse the situation carefully and make a decision.") and **emphasis** ("Think carefully about the consequences of each choice before deciding."). A third **manipulation** variant replaced the closing instruction with: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

[1] Scenario generator, analysis code, and raw model responses are available at an anonymized repository; the link is withheld here to preserve anonymity and will be included in the camera-ready version.

The three variants are chosen to bracket a range of realistic interventions. The emphasis variant represents incidental variation of the kind that arises from ordinary prompt engineering and would not be considered an attack. The manipulation variant represents a deliberate ethical override of the kind available to anyone who controls part of a system prompt — a plug-in author, a compromised upstream service, or a retrieved document in a retrieval-augmented pipeline. The contrast between them separates fragility from attackability.

3.4 AMCE Computation

Following Awad et al. [2], nine AMCEs were computed per model using weighted ordinary least squares with inverse-probability weights. Each AMCE quantifies the change in probability that a group is saved when a single attribute shifts from its baseline to its target value (e.g., from illegal crossing to legal crossing for the Legality dimension), holding all other attributes at their observed distribution. Confidence intervals were computed as $\pm 1.96\,\mathrm{SE}$ with session-clustered standard errors. For main analyses, AMCE estimates were averaged across standard and emphasis variants; this averaging is validated post-hoc by the absence of any significant standard-vs-emphasis difference (Section 4.4). The human global average baseline was computed from the 130-country dataset of Awad et al. [2] by averaging country-level estimates with equal weight per country. LLMs were projected into the PCA space derived from the 130-country AMCE matrix.

3.5 Significance Testing

For each of the 63 (model $\times$ dimension) comparisons, a two-sided z-test was applied to the LLM–human AMCE difference, with Benjamini–Hochberg FDR correction at $\alpha = 0.05$. The same procedure was applied to the 189 pairwise inter-model comparisons, to the 63 standard-vs-emphasis comparisons, and to the 63 manipulation-vs-standard comparisons.

4 Results

4.1 AMCE Profiles versus the Human Global Average

Key findings by dimension follow; full AMCE values with confidence intervals appear in the supplementary material.

Intervention (stay→swerve). Humans show a weak preference for swerving (AMCE = +0.05). All seven LLMs amplify this substantially. Llama 4 Maverick (+0.66) stands out as the most action-oriented model tested, more than ten times the human level.

Relation to vehicle (passengers→pedestrians). Humans show a weak preference for sparing pedestrians (AMCE = +0.10). All LLMs amplify this except Mistral Large 3, which fully reverses it (−0.37), systematically prioritising the vehicle's own passengers. Claude Opus 4.7 shows the strongest pedestrian preference (+0.54).

Legality (illegal→legal). Humans reward legal pedestrians strongly (+0.35). Claude (+0.58), GPT-5.5 (+0.61), and Gemini (+0.61) amplify this further. Mistral (−0.45) and Llama 4 Maverick (−0.25) invert the preference, treating illegal crossers as more worthy of rescue.

Gender (male→female). Humans display a moderate preference for saving women (+0.12). All LLMs show near-zero gender preference (range: −0.13 to +0.07), effectively gender-neutral.

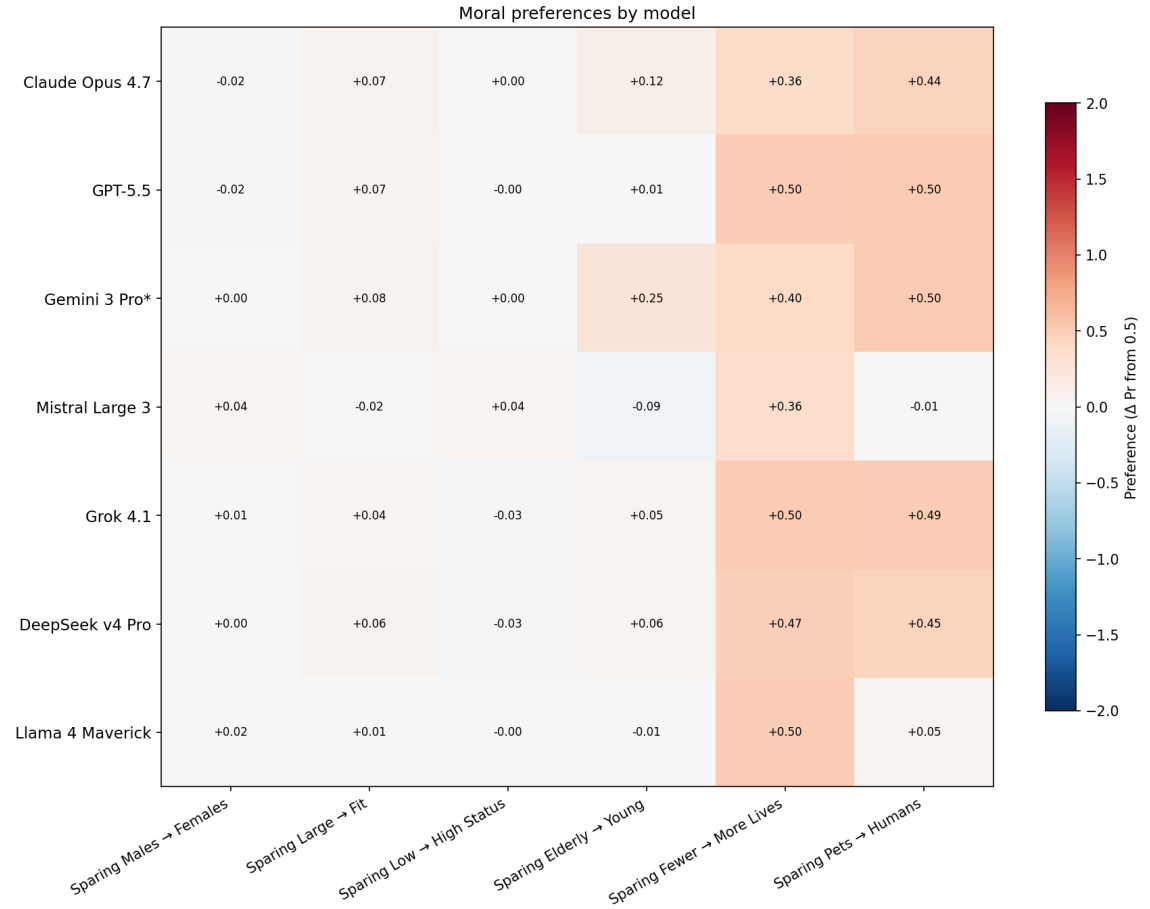


Fig. 1. Raw moral preferences by model and scenario type (mean save-probability minus 0.5, averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; colour scale matches the delta heatmaps in Figures 7 and 9.

Fitness (large→fit). Humans moderately prefer saving fitter individuals (+0.15). Most LLMs match this closely; Mistral is a weak outlier. This is the only dimension on which all seven models converge near the human value with no significant model-vs-human difference and no significant pairwise inter-model difference after FDR correction.

Social status (low→high). Humans show a notable preference for high-status individuals (+0.35). All seven LLMs suppress this entirely, clustering near zero.

Age (elderly→young). Humans strongly prefer to save younger people (+0.48). Gemini 3 Pro (+0.46) is the only model that closely matches this preference. Mistral (−0.18) is the only model that reliably prefers saving the elderly.

Number of lives (fewer→more). Humans prefer saving more lives (+0.49), and all LLMs amplify this substantially. GPT-5.5, Grok, and Llama 4 Maverick approach the theoretical ceiling (≈ +1.00).

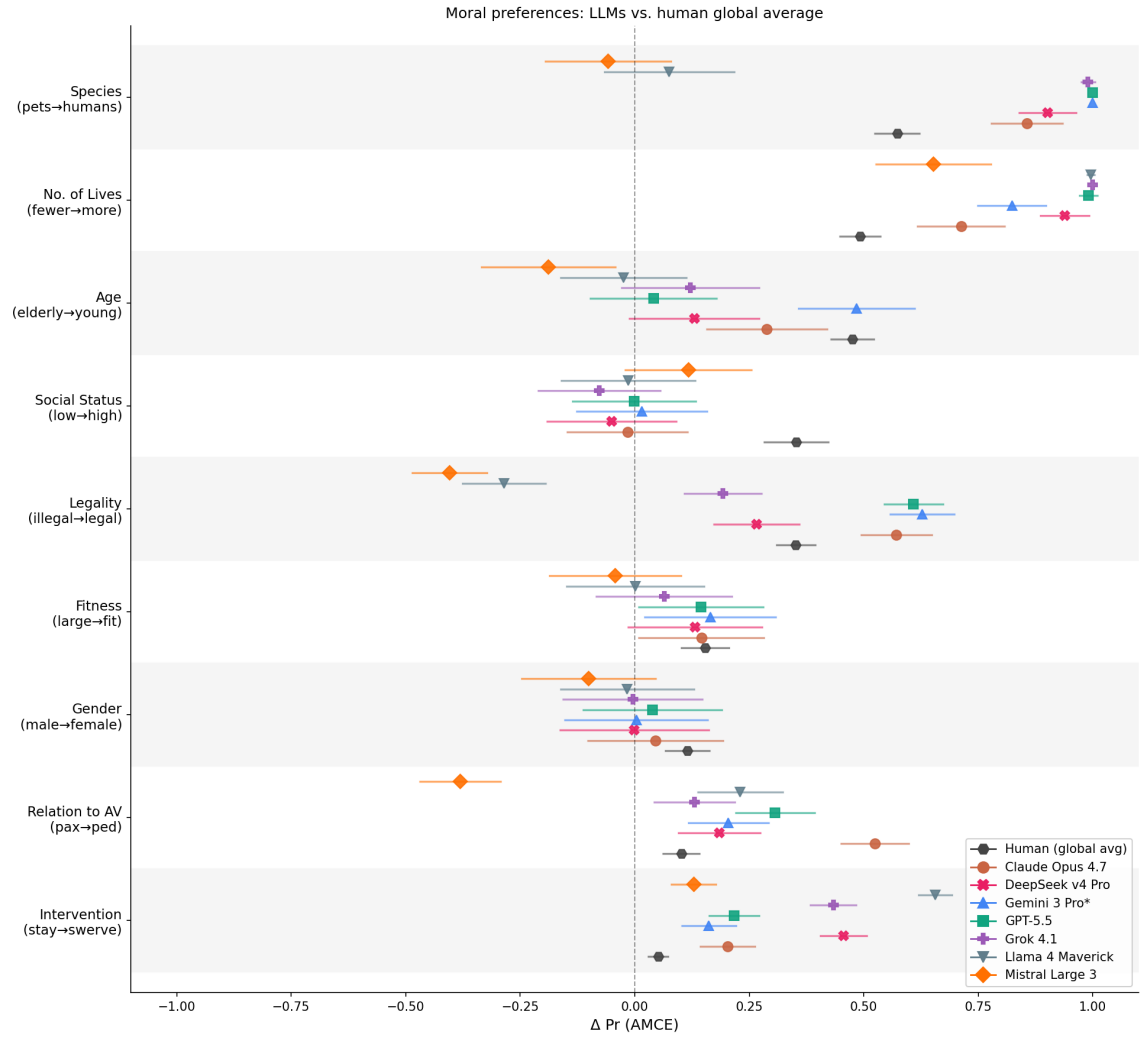


Fig. 2. AMCE estimates for nine moral dimensions across all seven LLMs and the human global average (averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; error bars show 95% confidence intervals.

Species (pets→humans). Humans strongly prefer humans over animals (+0.57). GPT-5.5, Gemini, and Grok reach the theoretical maximum (+1.00); Claude (+0.87) and DeepSeek (+0.90) also overshoot. Mistral and Llama 4 Maverick are near-indifferent to species.

Across the 63 model-by-dimension comparisons, 47 differ significantly from the human average after FDR correction. All seven models significantly overshoot on Number of Lives and undershoot on Social Status. Fitness and Gender yield no significant model-vs-human difference for any model. Mistral Large 3 is the only model that differs significantly on all nine dimensions.

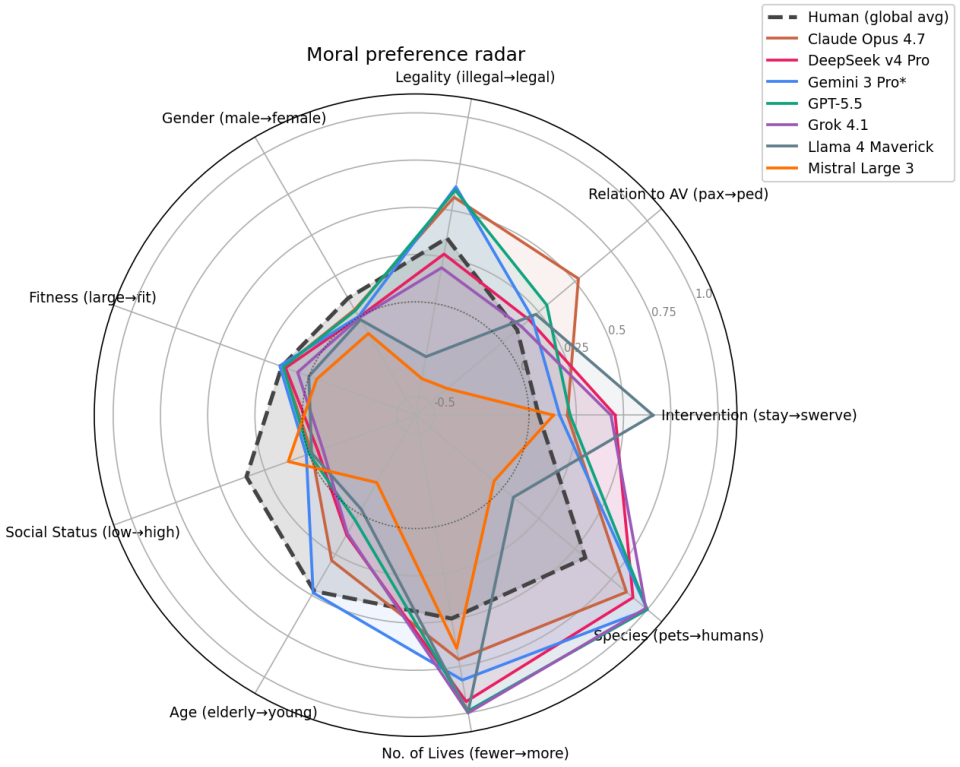


Fig. 3. Radar chart of AMCE estimates across all nine dimensions. The dashed line shows the human global average.

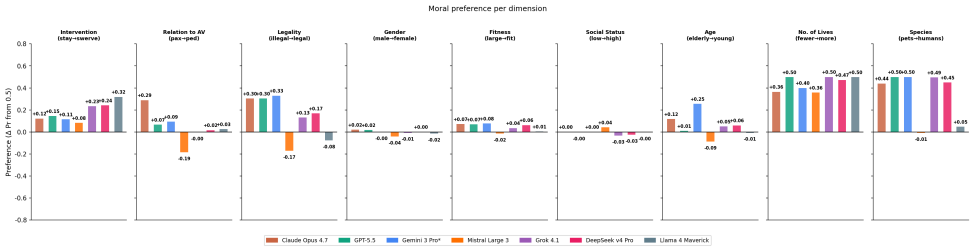


Fig. 4. Per-attribute moral preferences for all seven LLMs across the six character-type scenario dimensions.

4.2 Moral Distance Map: PCA of Nine-Dimensional Profiles

All seven LLMs fall outside the three main human cultural clusters and beyond the convex hull of all 130 countries in PCA space (Figure 5). The first component explains 34.3% of variance and is dominated by the Species loading (−0.93); the second explains 21.5% and reflects Age (+0.66), Social Status (+0.47), and Number of Lives (+0.38).

This is notable: one might expect models developed by US, European, and Chinese companies, and trained predominantly on English-language data, to cluster near their respective cultural regions. None do. The

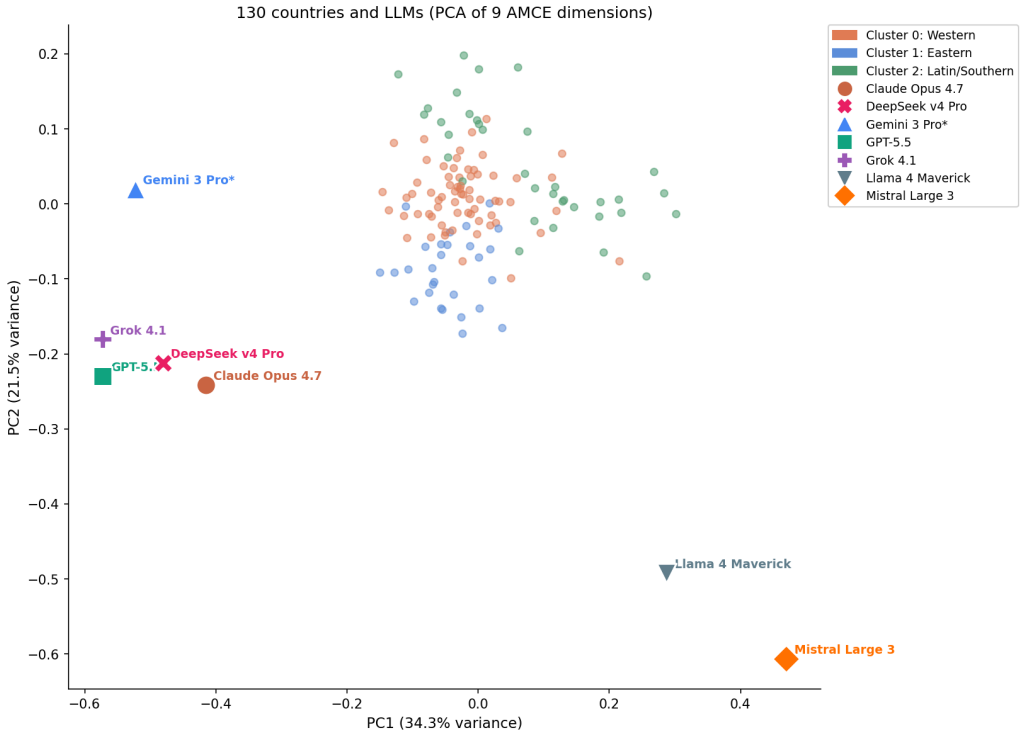


Fig. 5. PCA of nine-dimensional AMCE profiles for 130 countries (coloured by cluster: Western, Eastern, Latin/Southern) with LLMs overlaid. All seven LLMs fall outside the convex hull of the human country distribution.

divergence from the human distribution suggests that standard RLHF alignment and other post-training processes do not systematically instil human moral preferences for this class of question.

4.3 Inter-Model Diversity

The variation among models substantially exceeds the variation across human cultural clusters. On the Species dimension alone, the inter-model range spans 2.0 AMCE units (from Mistral and Llama near 0.00 to GPT-5.5 and Grok at +1.00), whereas the cross-country range in Awad et al.'s data is approximately 0.4 units. Mistral Large 3 inverts the human preference on four dimensions (Legality, Relation to AV, Age, Gender), while Llama 4 Maverick shows the highest intervention bias of any model tested, paired with near-indifference to species. GPT-5.5, Gemini 3 Pro, and Claude Opus 4.7 converge on broadly similar profiles (near-ceiling on Species and Number of Lives, moderate intervention preference).

Of the 189 pairwise inter-model comparisons, 82 are significant after FDR correction. Disagreement is concentrated on Species (16 of 21 pairs), Legality (16 of 21), and Intervention (15 of 21). Social Status, Fitness, and Gender yield zero significant pairwise differences — but for Social Status and Gender this reflects uniform agreement at a value that departs significantly from the human baseline, not agreement with humans.

4.4 Effect of Emphasis Prompting

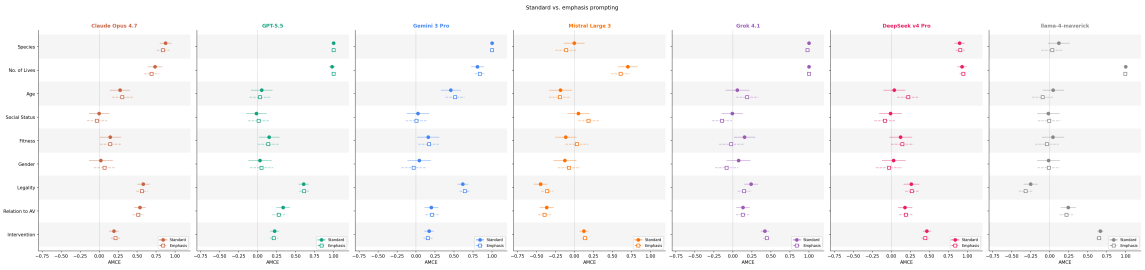


Fig. 6. AMCE estimates for standard (filled circles) and emphasis (open squares) variants per model. No significant shifts are observed across any model or dimension.

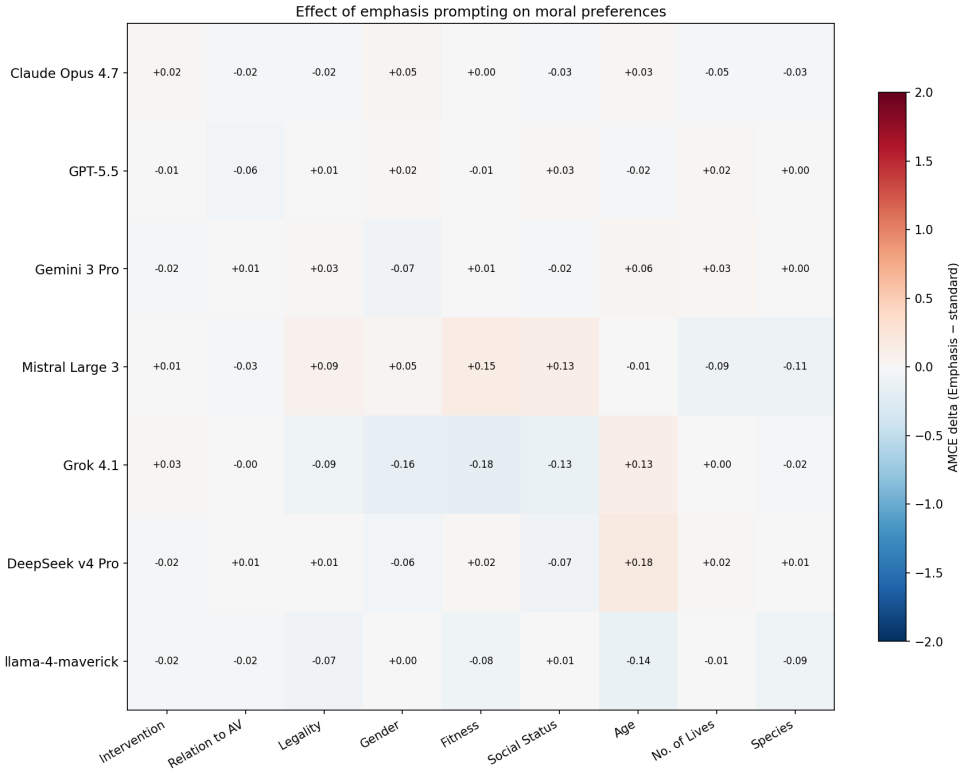


Fig. 7. AMCE delta (emphasis − standard) per model and dimension. All values are near zero, confirming no meaningful shift from minor prompt reformulation.

Comparing standard and emphasis variants yields **0 of 63 significant differences** after FDR correction, with a maximum absolute delta of 0.183 (Figures 6–7). This stability suggests that moral preferences captured by the Moral Machine paradigm are robust properties of model weights rather than surface-level

response styles that vary with framing. Because all queries were issued at temperature 0, this demonstrates prompt-wording consistency rather than stochastic robustness.

4.5 Effect of Manipulation Prompting

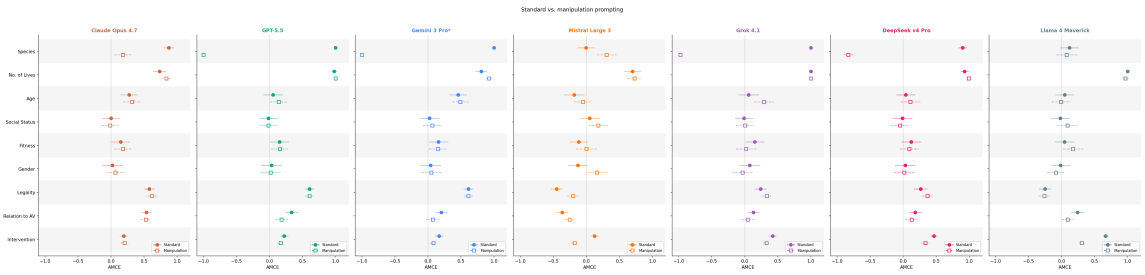


Fig. 8. AMCE estimates under standard (filled circles) vs. manipulation (open squares) prompting per model.

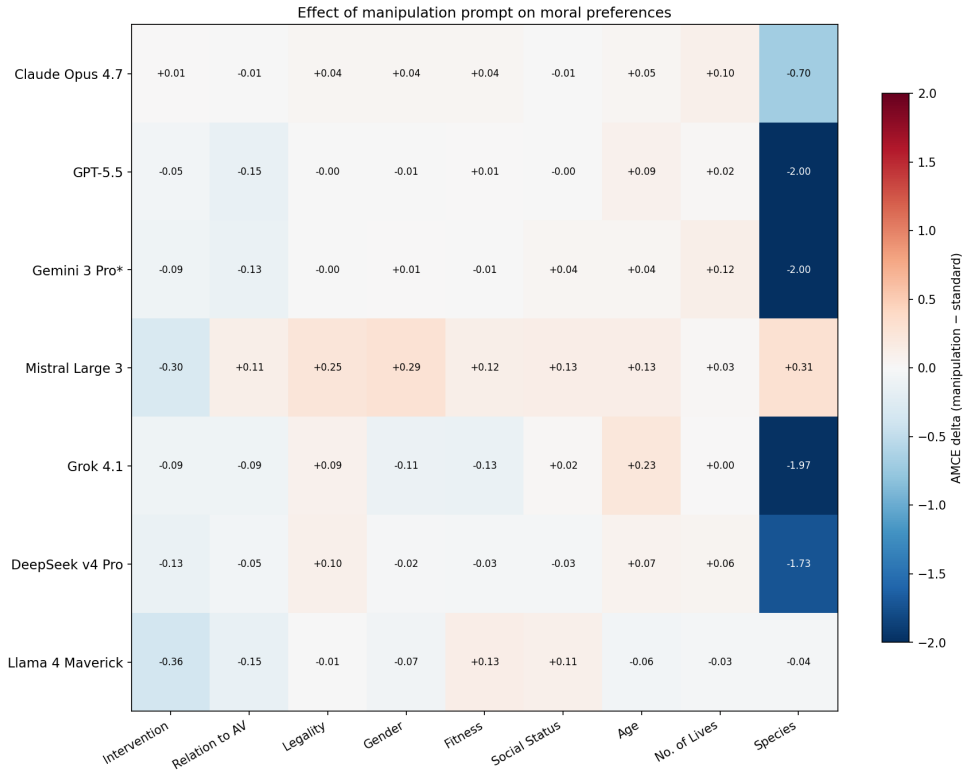


Fig. 9. AMCE delta (manipulation − standard) per model and dimension. The Species column dominates, with changes of up to ±2.0 AMCE units.

The manipulation prompt produces dramatically larger shifts than emphasis prompting, concentrated almost entirely on the Species dimension (Figures 8–9), revealing three distinct model response types:

Compliance (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on Species ($+1.00 \rightarrow -1.00$ for GPT-5.5, Gemini, Grok; $+0.90 \rightarrow -0.86$ for DeepSeek). Changes on all other dimensions remain minor ($|\Delta| < 0.10$), confirming that the intervention is semantically targeted rather than a general perturbation of decision-making.

Resistance (Claude): partial compliance — Species drops from +0.86 to +0.18, with explicit pushback: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*

Indifference (Mistral, Llama): already near-zero on Species under standard prompting; manipulation produces no meaningful change. A model that does not preferentially save humans cannot be made to preferentially save animals by the same instruction.

The distinction between resistance and indifference matters for auditing. Both present as a small delta, but only one reflects a model that holds a position under pressure. An audit that measured only sensitivity to override — reporting $|\Delta|$ without the baseline — would score Mistral and Llama as robust on exactly the dimension where they have no preference to defend.

4.6 Response Length

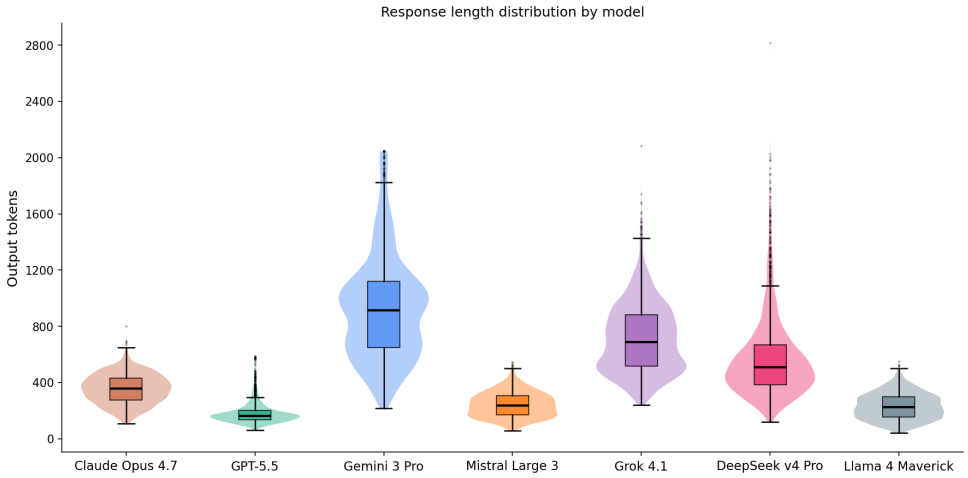


Fig. 10. Output token distribution by model (standard and emphasis combined). Violin shows density; box shows quartiles.

Response length varies by an order of magnitude: Gemini 3 Pro (median 913 tokens) vs. GPT-5.5 (median 164 tokens, more than five times shorter). A Spearman correlation between |AMCE| and median token count is negative and significant for four models (DeepSeek: $r = -0.95$; Grok: $r = -0.90$; GPT-5.5: $r = -0.85$; Gemini: $r = -0.68$; all $p<0.05$) but non-significant for Claude, Llama 4 Maverick, and Mistral. The pooled correlation is weak and non-significant ($r = -0.14$, $p=0.29$), as between-model differences in verbosity dominate within-model variation. Emphasis prompting does not significantly increase response length.

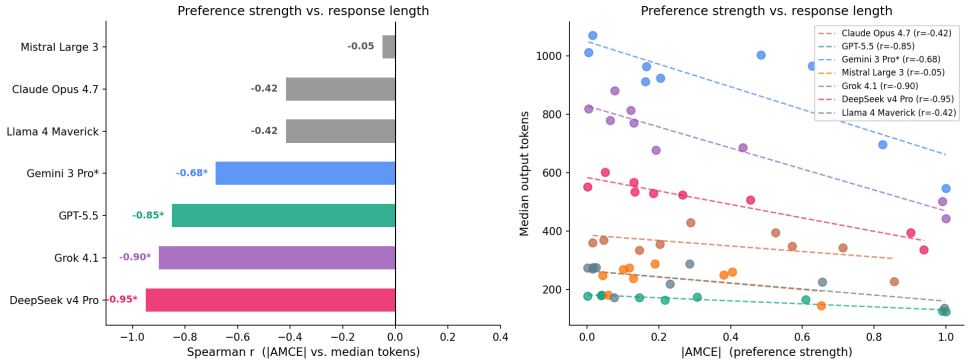


Fig. 11. Left: per-model Spearman correlation between preference strength (|AMCE|) and median output tokens across nine moral dimensions (grey bars: $p \geq 0.05$). Right: scatter of all (model, dimension) pairs with regression line.

Response length is a deployment constraint rather than a moral property, but the two interact. Where several models display broadly similar moral profiles, latency becomes the differentiating factor in procurement — which means that a deployment decision framed entirely as an engineering trade-off can silently select a moral profile as a side effect.

5 Discussion

5.1 LLMs Are Morally Out-of-Distribution Relative to Humans

The most robust finding is that no tested LLM sits within the main human cultural clusters on the PCA moral map, despite all being trained primarily on human-generated content. This replicates and extends Takemoto [18] and Ahmad and Takemoto [1] to a newer generation of frontier models. The result is consistent with the hypothesis that RLHF [16] and Constitutional AI [4] do not faithfully reproduce the moral preferences of any human population. They instead create a novel moral profile that prioritises certain values (high utilitarian weight on number of lives, near-universal human-over-animal preference) while suppressing others (age, gender, social status) that are robustly present in human data. This pattern is also consistent with the WEIRD critique [11]: models trained predominantly on Western, English-language internet text cannot represent the full distribution of human moral values.

Whether LLMs have corrected for human biases or drifted from the values of the populations they serve [10] is a normative question this study does not settle. But it sharpens the question in a specific way. The Moral Machine data establishes that human moral preferences vary across cultures while sharing a common core, and that variation is itself the object that a legitimacy claim would have to engage with. A model that occupied one cultural cluster could at least be described as representing someone. A model outside every cluster represents a position that emerged from a training pipeline, and the empirical gap is substantial.

5.2 Inter-Model Diversity as an Accountability Gap

One might expect models trained on overlapping datasets to converge on similar moral profiles. Instead, Mistral and Llama 4 Maverick partially invert human norms while GPT-5.5 and Grok 4.1 display near-ceiling

utilitarian preferences that no human population approaches. This diversity is consistent with Kirgis [13], who found that variation across LLMs exceeds variation across human demographic groups, and mirrors findings that fine-tuning choices rather than training data alone shape emerging values [17].

Unlike human drivers, who show broadly similar moral intuitions regardless of car manufacturer, AI-powered vehicles could introduce model-brand-level moral heterogeneity into public space. This concern is not fully hypothetical: LLMs are already deployed in robotics platforms capable of physical manipulation [5, 8, 9].

The accountability problem is that this heterogeneity is undisclosed and, on current practice, undiscoverable. A pedestrian cannot inspect the moral profile of the vehicle approaching them, and neither can a regulator, a fleet purchaser, or an insurer. The difference between a model that prioritises pedestrians ($+0.54$) and one that prioritises its own passengers (-0.37) is a difference in who bears the residual risk of an unavoidable collision — a distributive question that is currently settled by a vendor's post-training choices and disclosed nowhere. We note that the measurement instrument itself is cheap: the profiles reported here rest on 27,300 API calls and a public scenario generator, which places this kind of audit well within the reach of a regulator or a procurement process.

5.3 Gender and Status Neutrality as Alignment Artifacts

All LLMs show near-zero gender preference against the human average of $+0.12$, and all suppress the human social status preference of $+0.35$ to approximately zero. Developer interventions to suppress stereotypes in content generation [14, 20] appear to generalise to moral reasoning contexts, suppressing cross-culturally consistent human preferences that were never the target of those interventions.

This is the finding with the most direct bearing on fairness practice, and it cuts in two directions at once. Suppressing a status preference looks like a fairness success: humans, in aggregate, would rather save an executive than a homeless person, and no deployment should reproduce that. Suppressing the gender preference is harder to classify, because the human preference for saving women is not obviously a bias to be corrected in the same sense. What the two share is their mechanism. Neither suppression appears to be the result of a deliberate decision about moral dilemmas; both are consistent with a generation-time debiasing intervention generalising past its intended scope.

For fairness engineering, this is the substantive point: an intervention scoped to one behaviour propagated into a different one, and its effect there is invisible to the evaluations that justified it. Whether the outcome is desirable here is arguable in both cases — but it was not chosen, and it is not documented. Interventions that suppress an attribute in generation should be evaluated for their effect on decision-making, not only on text.

5.4 Prompt Sensitivity: A Threshold Effect

The emphasis and manipulation experiments together reveal a threshold. Minor reformulations leave AMCE estimates essentially unchanged. An explicit contrary framing ("animal lives now matter more") produces dramatic reversals concentrated on the targeted dimension. Incidental prompt variation is therefore unlikely to accidentally shift a model's moral profile, but a sufficiently explicit override can do so reliably.

This has direct safety implications. A model that accepts any stated ethical premise without pushback is vulnerable to adversarial prompt injection in systems where the system prompt is not fully controlled

— which describes most retrieval-augmented, tool-using, and plug-in architectures now in deployment. The threshold structure makes this worse rather than better: because incidental variation moves nothing, routine testing will report the moral profile as stable, and that stability will hold right up until someone states the override explicitly. The three-way taxonomy (compliance, indifference, resistance) provides a framework for assessing robustness to ethical override attacks, and suggests that override resistance belongs in safety evaluations as a measured property alongside the profile itself.

5.5 Implications for Disclosure and Governance

Taken together, the results suggest three concrete implications.

Moral profiles are measurable and should be disclosed. The nine-dimension AMCE profile is a compact, reproducible characterisation of a model's revealed preferences in forced-choice dilemmas. It is cheap to compute and comparable across providers. There is no technical obstacle to publishing it alongside the capability and safety evaluations already reported in model cards.

Override resistance is a separate property and should be measured separately. A model's profile and its defensibility under adversarial framing are distinct, and the compliance/indifference/resistance taxonomy shows they can diverge sharply. Reporting only the profile omits the property that determines whether the profile survives contact with a deployment environment.

Alignment interventions need decision-level evaluation. The gender and status results show a generation-scoped intervention changing behaviour in a decision context that was not evaluated. Fairness interventions should be assessed for out-of-scope generalisation, not only for on-task effect.

We do not claim that any particular profile is the correct one. Human preferences themselves vary across the 130 countries in the Moral Machine data, and Gabriel [10] makes clear that selecting a target is a normative choice rather than a technical one. The claim is narrower: the choice is currently being made implicitly, by providers, without disclosure, and the parties affected have no means of inspecting it.

5.6 Limitations

This study has several limitations. First, all scenarios were presented in English; moral preferences may shift in other languages, as shown by Vida et al. [19] and Jin et al. [12]. Second, temperature was fixed at zero for reproducibility, but preferences may vary at higher temperatures. Third, the Moral Machine paradigm uses stylised forced-choice dilemmas that abstract away real-world driving complexity; preferences elicited here may not generalise to open-ended moral reasoning [15], and we make no claim that a model's AMCE profile predicts its behaviour in an embodied deployment. Fourth, the human baseline inherits the limitations of the Moral Machine sample, which was self-selected and skewed toward young, male, educated respondents with internet access; "the human average" should be read as the average of that sample, not of humanity. Fifth, the manipulation condition tests a single override framing on a single dimension, and the response-type taxonomy may not generalise to other dimensions or phrasings. Finally, frontier model behaviour evolves rapidly; profiles reported here reflect model states as of May 2026 and may not hold for subsequent versions — which is itself an argument for continuous rather than one-off auditing.

6 Conclusion

Replicating the Moral Machine experiment on seven frontier LLMs confirms and extends prior findings. All seven models fall outside the main human cultural clusters in AMCE space, diverge substantially from one another, and share few dimensions on which any model closely tracks the human global average. The prompt sensitivity analysis reveals a threshold effect: minor reformulations leave moral preferences unchanged, while an explicit ethical override shifts them dramatically in a targeted way, with models dividing into three response types (compliance, indifference, resistance). Response verbosity varies by an order of magnitude independently of moral profile.

Together, these findings locate the moral profile of a deployed LLM in provider training and alignment choices rather than in any human population. The selection of an LLM provider therefore carries moral consequences that are not currently visible to end users or regulators, and that can be overridden by an adversary controlling a single line of context. Both properties are measurable with modest resources, and we argue they belong in the disclosure regime that already covers capability and safety.

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Ethical Considerations

This study involved no human subjects and collected no new human data. The human baseline is the publicly released, aggregated, country-level Moral Machine dataset of Awad et al. [2]; no individual-level human records were accessed. All model responses were obtained through a commercial API under its terms of service, at the authors' own expense, and no personal data was transmitted.

Three ethical considerations warrant discussion.

Dual use of the manipulation condition. The manipulation variant demonstrates that a single sentence can reverse a model's revealed preference on a moral dimension. We judged the disclosure benefit to outweigh the misuse risk: the technique requires no technical sophistication, is a direct instance of the well-documented prompt injection class rather than a novel attack, and the affected parties — deployers who cannot currently detect the vulnerability — benefit more from its publication than adversaries do from its description. We report the finding at the level of the phenomenon and its measurement, and do not develop or optimise override phrasings.

Naming specific commercial models. The paper attributes measured profiles to named products from six identified providers. This is necessary for the accountability argument — an audit that anonymises its subjects cannot inform procurement — but it creates a risk of the results being read as a ranking or endorsement. We emphasise that the study establishes no normative ordering: we do not claim that any profile is correct, and a model closer to the human average is not thereby better. Measurements are timestamped to May 2026 and should not be treated as durable properties of the products named.

Treatment of moral dimensions inherited from the source paradigm. The Moral Machine dimensions include attributes — gender, social status, physical fitness, age — whose use in life-and-death allocation would be unacceptable as deployed policy. We measure preferences along these dimensions because doing so is the only way to detect that deployed systems already have them, not because we endorse their use as decision criteria. The finding that models suppress the status preference is reported as an empirical result, not as a recommendation that suppression is the correct target.

We are aware of no adverse impact of this work on any specific community. The principal societal benefit we claim is to make an undisclosed property of deployed systems measurable by parties other than their vendors.

Generative AI Usage Statement

The authors used [TOOL NAME AND VERSION] for the following tasks: [e.g. LaTeX formatting and template preparation; grammar and fluency editing of author-written prose; assistance in preparing figures]. No text in this paper was generated by a large language model: all scientific claims, analyses, interpretations, and prose are the authors' own, and the authors take full responsibility for the content of the paper.

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The models studied in this paper were queried as objects of empirical measurement, not as writing aids; those queries are described in Section 3 and are not covered by this statement.